2019 Ph H2 Q10

Section: Particles and Waves

Topic: Wave-Particle Duality

A beam of electrons is passed through a narrow gap and produces a diffraction pattern on a screen.

(a)
Explain why a diffraction pattern is produced

✅ According to the **wave-particle duality**, particles like electrons can behave like waves.
When electrons pass through a **narrow gap** comparable in size to their **wavelength**, they **diffract**, producing a **pattern of maxima and minima** — just like light or water waves.

(b)
State what is observed in the diffraction pattern when electrons of higher energy are used

✅ Higher energy electrons have a **shorter wavelength** (from the de Broglie relation).
As a result, the diffraction pattern becomes **less spread out** — the **rings get closer together**.

(c)
Explain this observation using the de Broglie equation

The **de Broglie wavelength** is:

$$\lambda = \frac{h}{p}$$

Where:

- h is Planck's constant
- $p = mv$ is the momentum of the electron

When electrons gain more energy, their speed (and hence momentum) increases.
As p increases, λ decreases, so:

✅ **Shorter wavelength → narrower diffraction pattern**

🧠 Revision Tips

- **Electron diffraction** is key evidence for wave–particle duality
- The **narrower the gap**, the more pronounced the diffraction
- Higher energy → higher momentum → smaller λ → less spreading
- The de Broglie equation:

$$\lambda = \frac{h}{mv}$$

applies to all matter particles
